

1.) Assigned 3

$$\frac{4^n - 1}{3} \text{ if } n = k \text{ then } \frac{4^k - 1}{3} \text{ for } n = 1 \quad \frac{4^1 - 1}{3} = 1$$

∴ the base case holds

$$\text{if } n = k+1 \text{ then } \frac{4^{k+1} - 1}{3} = \frac{4 \cdot 4^k - 1}{3} = \frac{4 \cdot 4^k - 4 + 4 - 1}{3} = \frac{4(4^k - 1) + 3}{3} = \frac{4(4^k - 1)}{3} + 1$$

∴ by mathematical induction the statement holds

$$2.) a_1 = 4 \quad a_2 = 12 \quad a_n = 4a_{n-1} - 4a_{n-2} \text{ if } n \geq 3$$

use induction for $a_n = 2^n(n+1)$

Base case: if $n = 1$ ∴ $2^1(1+1) = 4$ ✓ ∴ the base case holds true

if $n = 2$ ∴ $2^2(2+1) = 12$ ✓ holds true

the assume $S(m)$ is true for all integers m when $1 \leq m \leq k$ to show $S(k+1)$ is true.

$$\text{for } a_{k+1} = 4a_k - 4a_{k-1}$$

$$a_{k+1} = 2^{k+1}(k+1) \cdot 4 - 2^{k+1}(k) \cdot 4$$

$$\hookrightarrow 2^{k+1}(2k+2) - 2^{k+1}(k)$$

$$\hookrightarrow 2^{k+1}(k+2)$$

3a.) for contradiction the we only needed to say case 1, as if both $a < 5$ and $b < 5$ then it can never be greater than or equal to nine in $\mathbb{Z}^+$.

for contra position if it is claimed $a+b < 10 \rightarrow a+b < 9$ which is not true

b.) the negation of $a \geq 5$, $a \leq 4$, ∴ as the statement is $a \geq 5 \vee b \geq 5$, both need to be negated. ∴ the statement become $a \leq 4$ and $b \leq 4$, then the equation of $a+b \leq 8$ shows that $a+b$ cannot ≥ 9 .